

Data Flow Diagram

Date	08/07/2026
Team ID	
Project Name	Human Development Index Predictor
Maximum Marks	2 Marks

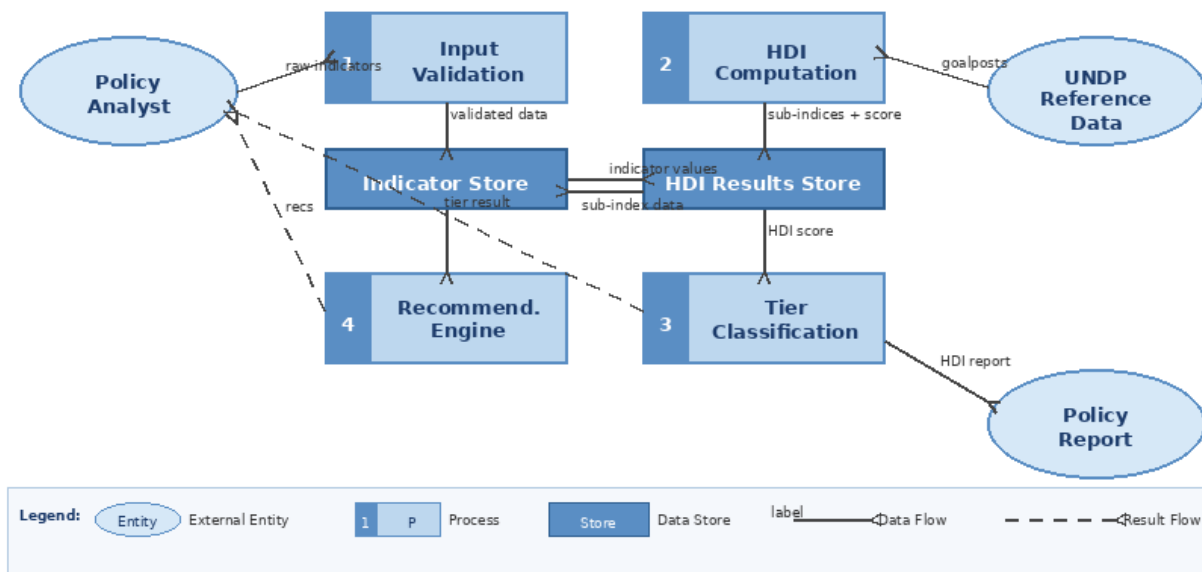
Data Flow Diagram

Create a Data Flow Diagram (DFD) to show how data moves through your system — between external entities, processes, and data stores. Use the symbol legend below as a guide.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	A person, organization, or system outside the project that sends or receives data (e.g., Policy Analyst, UNDP Reference).
Rectangle with numbered header	Process	An activity that transforms incoming data into outgoing data (e.g., Input Validation, HDI Computation).
Rectangle (solid fill, no number)	Data Store	A place where data is held for later use (e.g., Indicator Store, HDI Results Store).
Labeled arrow	Data Flow	The movement of data between entities, processes, and data stores, labeled with what data is being passed.

HDI Predictor — Data Flow Diagram



DFD Description

Node	Name	Description
External Entity	Policy Analyst	The primary user who inputs life expectancy, schooling years, and GNI per capita and receives HDI tier results and recommendations.
External Entity	UNDP Reference	Provides the official min–max goalposts used to normalise each indicator into a sub-index value (e.g. life expectancy: 20–85 years).
External Entity	Policy Report	The output destination — a ministry or research body that receives the exported HDI result as a PDF or CSV report.
Process 1	Input Validation	Receives raw indicator values from the analyst. Validates ranges (e.g. GNI \$100–\$75,000) and rejects out-of-bound inputs before passing clean data to the Indicator Store.
Process 2	HDI Computation	Reads validated indicator values from the Indicator Store and the UNDP goalposts. Computes health, education, and income sub-indices, then calculates the composite HDI score using the geometric mean formula.
Process 3	Tier Classification	Reads the HDI score from the HDI Results Store and maps it to the correct tier: Very High (≥ 0.800), High (0.700–0.799), Medium (0.550–0.699), or Low (< 0.550). Returns the tier result to the analyst and sends a report to the Policy Report entity.
Process 4	Recommendation Engine	Reads the sub-index breakdown from the HDI Results Store. Identifies the lowest-scoring dimension (health, education, or income) and generates targeted policy recommendations returned to the analyst.
Data Store 1	Indicator Store	Stores the validated raw indicator inputs submitted by the analyst — life expectancy, mean years of schooling, expected years of schooling, and GNI per capita.
Data Store 2	HDI Results Store	Stores computed sub-index values (health, education, income) and the composite HDI score produced by the HDI Computation process for use by Classification and Recommendation processes.